25th

Dictionary

In [3]:
```
# Dictionary is a mutable data type in Python.
# A python dictionary is a collection of key and value pairs separated by a colo
# in curly braces {}.
# Keys must be unique in a dictionary, duplicate values are allowed.
```

Create Dictionary

In [4]:
```
mydict = {} # Empty dictionary
mydict
```

Out[4]:
```
{}
```

In [5]:
```
type(mydict)
```

Out[5]:
```
dict
```

In [6]:
```
mydict = {1:'one', 2:'two', 3:'three'} # dictionary with integer keys
mydict
```

Out[6]:
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [7]:
```
mydict = dict({1:'one' , 2:'two', 3:'three'})  # Create dictionary using dict()
mydict
```

Out[7]:
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [10]:
```
mydict = {'A' :'one',  'B':'two', 'C':'three'}  # dictionary with character keys
mydict
```

Out[10]:
```
{'A': 'one', 'B': 'two', 'C': 'three'}
```

In [11]:
```
mydict = {1:'one' , 'A':'two' , 3:'three'} # dictionary with mixed keys
mydict
```

Out[11]:
```
{1: 'one', 'A': 'two', 3: 'three'}
```

In [12]:
```
mydict.keys() # Return Dictionary Keys using keys() method
```

Out[12]:
```
dict_keys([1, 'A', 3])
```

In [13]:
```
mydict.values() # Return Dictionary values using values() method
```

Out[13]:
```
dict_values(['one', 'two', 'three'])
```

In [14]:
```
mydict.items() # Access each key-value pair within a dictionary
```

Out[14]: dict_items([(1, 'one'), ('A', 'two'), (3, 'three')])

In [16]:
```
mydict = {1:'one', 2:'two', 'A':{'Name':'asif' , 'Age': 25} , 'B': {'Bat' , 'Cat
mydict
```

Out[16]:
```
{1: 'one',
 2: 'two',
 'A': {'Name': 'asif', 'Age': 25},
 'B': {'Bat', 'Cat', 'Hat'}}
```

In [17]:
```
keys = {'A' , 'B' , 'C' , 'D'}
mydict3 = dict.fromkeys(keys) # Create a dictionary from a sequence of keys
mydict3
```

Out[17]: {'D': None, 'B': None, 'A': None, 'C': None}

In [18]:
```
keys = {'A' , 'B' , 'C' , 'D'}
value = 10
mydict3 = dict.fromkeys(keys, value) # Create a dictionary from a sequence of ke
mydict3
```

Out[18]: {'D': 10, 'B': 10, 'A': 10, 'C': 10}

In [20]:
```
keys = {'A' , 'B' , 'C' , 'D'}
value = {10, 20 , 30}
mydict3 = dict.fromkeys(keys, value) # Create a dictionary from a sequence of ke
mydict3
```

Out[20]: {'D': {10, 20, 30}, 'B': {10, 20, 30}, 'A': {10, 20, 30}, 'C': {10, 20, 30}}

In [25]:
```
value.append(40)
mydict3
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[25], line 1
----> 1 value.aad(40)
      2 mydict3

AttributeError: 'set' object has no attribute 'aad'
```

Accessing Items

In [26]:
```
mydict = {1: 'one', 2:'two' , 3:'three' , 4:'four'}
mydict
```

Out[26]: {1: 'one', 2: 'two', 3: 'three', 4: 'four'}

In [27]:
```
mydict[1] # Access item using key
```

Out[27]: 'one'

In [28]:
```
mydict.get(1) # Access item using get method
```

Out[28]: 'one'

In [35]:
```
mydict1 = {'Name' :'Mohit' , 'ID': 1234, 'DOB':2000, 'Job':'Data Analyst'}
mydict1
```

Out[35]:
```
{'Name': 'Mohit', 'ID': 1234, 'DOB': 2000, 'Job': 'Data Analyst'}
```

In [38]:
```
mydict1['Name'] # Access item using key
```

Out[38]:
```
'Mohit'
```

In [39]:
```
mydict.get('Job') # Access item using get method
```

Out[39]:
```
'Data Analysist'
```

Add , Remove & Change items

In [40]:
```
mydict1 = {'Name':'Arif' , 'ID': 25 , 'DOB': 1993 , 'Address':'Hilsini'}
mydict1
```

Out[40]:
```
{'Name': 'Arif', 'ID': 25, 'DOB': 1993, 'Address': 'Hilsini'}
```

In [42]:
```
mydict1['DOB'] = 1992 # Changing Dictionary Items
mydict1['Address'] = 'Delhi'
mydict1
```

Out[42]:
```
{'Name': 'Arif', 'ID': 25, 'DOB': 1992, 'Address': 'Delhi'}
```

In [43]:
```
dict1 = {'DOB':1995}
mydict1.update(dict1)
mydict1
```

Out[43]:
```
{'Name': 'Arif', 'ID': 25, 'DOB': 1995, 'Address': 'Delhi'}
```

In [46]:
```
mydict1['Job'] = 'Analyst' # Adding items in the dictionary
mydict1
```

Out[46]:
```
{'Name': 'Arif', 'ID': 25, 'DOB': 1995, 'Address': 'Delhi', 'Job': 'Analyst'}
```

In [47]:
```
mydict1.pop('Job') # Removing items in the dictionary using Pop method
mydict1
```

Out[47]:
```
{'Name': 'Arif', 'ID': 25, 'DOB': 1995, 'Address': 'Delhi'}
```

In [48]:
```
mydict1.popitem() # A random item is removed
mydict1
```

Out[48]:
```
{'Name': 'Arif', 'ID': 25, 'DOB': 1995}
```

In [50]:
```
del[mydict1['ID']] # Removing items using del method
mydict1
```

Out[50]:
```
{'Name': 'Arif', 'DOB': 1995}
```

In [51]:
```
mydict1.clear() # Delete all items of the dictionary using clear method
mydict1
```

Out[51]: {}

In [52]:
```
del mydict1 # Delete dictionary object
mydict1
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[52], line 2
      1 del mydict1 # Delete dictionary object
----> 2 mydict1

NameError: name 'mydict1' is not defined
```

Copy Dictionary

In [53]:
```
mydict = {'Name':'Arif', 'ID':123 , 'DOB':2005 , 'Address':'Hilsini'}
mydict
```

Out[53]: {'Name': 'Arif', 'ID': 123, 'DOB': 2005, 'Address': 'Hilsini'}

In [54]:
```
mydict1 = mydict # Create a new reference "mydict1"
```

In [55]:
```
id(mydict) , id(mydict1)  # The address of both mydict & mydict1 will be the sam
```

Out[55]: (2323738602880, 2323738602880)

In [56]:
```
mydict2 = mydict.copy() # Create a copy of the dictionary
```

In [57]:
```
id(mydict2)  # The address of mydict2 will be different from mydict because mydi
```

Out[57]: 2323738122304

In [58]:
```
mydict['Address'] = 'Mumbai'
mydict
```

Out[58]: {'Name': 'Arif', 'ID': 123, 'DOB': 2005, 'Address': 'Mumbai'}

In [59]:
```
mydict1 # mydict1 will be also impacted as it is pointing to the same dictionary
```

Out[59]: {'Name': 'Arif', 'ID': 123, 'DOB': 2005, 'Address': 'Mumbai'}

In [60]:
```
mydict2  # Copy of list won't be impacted due to the changes made in the origina
```

Out[60]: {'Name': 'Arif', 'ID': 123, 'DOB': 2005, 'Address': 'Hilsini'}

Loop through a Dictionary

In [62]:
```
mydict1 = {'name':'Arif' , 'ID':123 , 'DOB':2007, 'Address':'Mumbai'}
mydict1
```

Out[62]: {'name': 'Arif', 'ID': 123, 'DOB': 2007, 'Address': 'Mumbai'}

In [64]:
```
for i in mydict1:
     print(i , ':' , mydict1[i]) # Key & value pair
```

```
name : Arif
ID : 123
DOB : 2007
Address : Mumbai
```

In [65]:
```
for i in mydict1:
    print(mydict1[i]) # Dictionary items
```

```
Arif
123
2007
Mumbai
```

Dictionary Membership

In [66]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Job': 'Analyst'}
mydict1
```

Out[66]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Job': 'Analyst'}
```

In [68]:
```
'Name' in mydict1 # Test if a key is in a dictionary or not
```

Out[68]:
```
True
```

In [72]:
```
'Asif' in mydict1 # Membership test can be only done for keys.
```

Out[72]:
```
False
```

In [70]:
```
'ID' in mydict1
```

Out[70]:
```
True
```

In [71]:
```
'Address' in mydict1
```

Out[71]:
```
False
```

All / Any

The All() method returns:

In [ ]:
```
# True - If all all keys of the dictionary are true
# False - If any key of the dictionary is false
#The any() function returns True if any key of the dictionary is True. If not, a
```

In [73]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Job': 'Analyst'}
mydict1
```

Out[73]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Job': 'Analyst'}
```

In [74]:
```
all(mydict1) # Will Return false as one value is false (Value 0)
```

Out[74]:
```
True
```

In [ ]:
```

```